

AMAL FATEMAH

Game Researcher & Developer

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ABOUT ME

Game researcher and developer with 5+ years of experience creating immersive experiences with Unity. Currently a doctoral researcher studying player experience and sustainability in games. Driven by curiosity, a problem-solving mindset and a passion for lifelong learning.

WORK EXPERIENCE

Doctoral Researcher (Game Design & Player Research) at LUT, Finland Aug 2024 – Present

- Conducted comprehensive literature review on use of games in sustainability education.
- Designed, administered and analyzed survey data to collect insights on nature representation and perceptions in games.
- Performed exploratory playthroughs of mobile games to evaluate mechanics and user experience.

Team Lead | Senior Developer (Games) at Artrei Studio (Remote) May 2023 – Aug 2026

- Built a focus-training game for children and integrated live heart rate monitoring.
- Led blockchain strategy game development while implementing in-game NFT minting and transfers.
- Oversaw multiplayer NFT card game using Photon and Firebase.
- Directed F2P green game activations backed by research insights.

QA Tester at KuvaLearn (AI-based Finnish Language Learning App) (Volunteer) Aug 2025 – Present

- Conducted systematic testing across mobile and web platforms to identify and document usability issues.
- Evaluated AI-assisted learning features and provide UX feedback to improve experience and interaction flows.
- Collaborated with developers to reproduce issues and improve product usability.

Intelgency IT Solutions (Pvt) Ltd, Pakistan

- Senior Unity Developer
- Unity Developer

Dec 2020 – April 2023

July 2021 – April 2023

Dec 2020 – July 2021

Achievements:

- Led development of multiple Metaverse play-to-earn entertainment platforms.
- Oversaw the integration of a game-based distance learning system while implementing core features.
- Developed an AR Game Show application and a Conversational AI AR Chatbot.
- Developed a Top-Down wave based survival shooting mining game.
- Worked on multiple live location AR mobile games using Mapbox and Google Maps.
- Published cross-platform mobile games, including iOS App Store releases.
- Provided technical mentorship and support to junior developers.

Project Management Trainee at Operations PRO (Pvt) Ltd, Pakistan Sep 2020 – Nov 2020

- Led a team of data analysts and graphic designers to publish a series of reports on Digital Business trends.
- Developed marketing strategies with a cross functional team to launch and promote a comprehensive CMS.

Coding Coach at That Lab, Pakistan Jan 2018 – March 2018

- Developed interactive, activity-based curricula of Robotics and IoT for students aged between 7 -14 years.
- Defined learning outcomes and managed STEM lab setup.

SKILLS

Research: Survey Design, Experimental Design, Literature Reviews, Research Design, Grant/Proposal Writing

Data & Analysis: Statistical Analysis (Regression, SEM), Unity Analytics, Excel, Stata, Python

Game Development & XR: Unity (2D/3D, AR, VR, Multiplayer), Gameplay Programming, UI/UX, XR Interaction Design, Basic 3D Modeling (Blender, 3D asset integration & optimization)

Project Management: Technical Documentation, SysML, Jira

Core Strengths: Agency, Creative Problem Solving, Adaptability, Systems Thinking, Cross-disciplinary collaboration

EDUCATION

PhD Engineering Science (Industrial Engineering & Management) 2024 - Present

Lappeenranta-Lahti University of Technology (LUT), Lahti, Finland

Dissertation Topic: *Designing for Planetary Play: Embedding Ecological Limits in Game Mechanics*

MS Systems Engineering (Specialization: Cognitive Systems) 2019

NUST, RCMS, Islamabad, Pakistan - (CGPA 3.95 / 4.00 - Gold Medallist)

Thesis Topic: *Interactive 3D Molecular Visualization for Spatial Learning*

BE Mechatronics - (CGPA 3.54 / 4.00) 2017

NUST, CEME, Rawalpindi, Pakistan

Final Year Project: *Design of an Assistive Robot for Domestic Applications*

SELECTED PROJECTS

- **Molecular Visualization Application** (MS Thesis Project) Unity (Android, Linux)
 - Multi-threaded client server model which converted 2D textbook images to 3D models.
 - Collected & analysed student data - results of which showed an increase in student test scores by up to 11%.
- **Schole** (Game-based distance learning system) Unity (Android, iOS, Web)
 - Multi-role system (Teacher, Student, Parent, Admin) with live class and whiteboard synchronization via Photon
 - Authentication and Data Management using PlayFab
- **Metabox** (AR Game Show) Unity (Android, iOS, Desktop)
 - Firebase backend with AR puzzle boxes; desktop admin tool for uploading puzzles images and videos
- **AkademiaAI** (AR Chatbot) Unity (Android, iOS)
 - Voice-enabled AR chatbot with IBM Watson Assistant + TTS/STT APIs.
- **Crypto Champions** (Block chain strategy game) Unity (Android, iOS, Web)
 - NFT minting via contracts and NFT hosting on Pinata.
 - NFT management, authentication, chats, user and opponent management using *Firebase Rest APIs*.
- **Micropets2D** (Multiplayer NFT card game, Pets Poker) Unity (Web)
 - Photon-based multiplayer; Firebase for authentication and data persistence
- **Remove This: Guns & Aliens** (Top-Down Shooting Game) Unity (Android, iOS)
 - Wave based survival shooting game with metamask integration for crypto rewards.
- **Lazy Horse Race Club** (Metaverse Entertainment World) Unity (VR - MetaQuest, Web, Desktop)
 - Firebase-backed multiplayer VR casinos, concerts and customizable apartments.
 - NFT Authentication on the Cronos chain
- **Firefight** (VR Fire Extinguisher Training) Unity (VR - MetaQuest)
 - PASS (Pull-Aim-Squeeze-Sweep) method training simulator for proper fire extinguisher use.
- **GeoFinding** (Play-to-Earn location based AR game) Unity (Android, iOS)
 - Firebase-integrated geolocation gameplay using Mapbox
- **Algocyte VR** (VR Experience) Unity (VR – MetaQuest, Apple Vision Pro)

- VR experience simulating how an AI based medical device sees and captures images in order to produce a test result.

▪ Others

- More projects and details: <https://amalfatemah.github.io/#projects>
- Play Store: <https://play.google.com/store/apps/dev?id=8061148689500685827>
- App Store: <https://apps.apple.com/us/developer/muhammad-arслан-touqeer/id1561452665>

RESEARCH FUNDING AND GRANTS

August 2026 – January 2027, Doctoral research funding, Suomen kulttuurirahasto, 16000€

August 2024 – April 2025, Doctoral research funding, EDUFI Fellowship, 17100€

July 2025 – September 2025, Doctoral research funding, EDUFI Fellowship, 5700€

May 2025, Travel funding for dissemination at Nordic DiGRA 2025, Finnish Society for Game Research

RESEARCH OUTPUT

Total publications: **4**, Total citations: **160**, h-index: **2** (Google Scholar)

Journal Articles:

Amal Fatemah, Shahzad Rasool, Uzma Habib. (2020). *Interactive 3D visualization of chemical structure diagrams embedded in text to aid spatial learning process of students*. Journal of Chemical Education. 97(4), <https://doi.org/10.1021/acs.jchemed.9b00690>

Mohammad Mohammad, Amal Fatemah, Lobna Hassan. (2024). *Effects of gamification on motivations of elementary school students: an action research field experiment*. Simulation & Gaming, <https://doi.org/10.1177/10468781241237389>

Minhal Fatemah, Amal Fatemah, Saima Jawad. (2026). *The SEARCH for Evidence: Designing a Game-Based Experiential Learning Intervention for Evidence-Based Medicine*. Medical Science Educator. <https://doi.org/10.1007/s40670-026-02671-1>

Conference:

Amal Fatemah, Lobna Hassan, Mirva Hyypiä. (2025). *Towards a Hopeful Future: A Literature Review of Gamified and Game-Based Approaches in Sustainability Education*. Proceedings of Nordic DiGRA 2025: Hope – Envisioning the Future of Game Cultures, <https://doi.org/10.26503/dl.v2025i1.2413>

OTHER EDUCATION AND EXPERTISE

March 2026, **Data Analytics with Python** and **Databases** courses, SOBIGDATA Research Infrastructure

26–28 August 2025, STRATEGIES Summer School: **Ecogames and Sustainability in the Video Game Industry** (Horizon Europe funded project), **University of Warsaw**, Poland.

Aug 4th, 2018 – Sep 15th, 2018, **Machine Learning** Hands-on Workshops, Experts Vision (EVEATI LTD), Pakistan.